

VISHAL RAJ

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SKILLS

Technical skills: Cloud Security, DevSecOps, Terraform IAC, Kubernetes, AWS, GCP, Hybrid Cloud, Docker, Python, Full Stack Web Development, Web Scraping, Oracle, MySQL, SQLite, Data Modeling and Design, Git, AI/ML, Data Analysis/Processing, Automation, HTML, CSS, JavaScript, C++, JAVA.

Functional Skills: Product Quality/Design, Testing, Project Planning/Timeline Development, Agile (Scrum, Kanban) and Waterfall Project Execution Methodologies, Data Strategy, Requirements Gathering, Algorithm Design.

Programming Language: Python

- Experienced in Python Libraries: Pandas, Numpy, TensorFlow, scikit-learn, Matplotlib, Seaborn, MatLabEngine, Flask, Zappa, Beautiful Soup, Selenium Webdriver, tkinter, Turtle Graphics.
- Studying and working with python for 6+ years.
- Certificates: 100 Days of Code: The Complete Python Pro Bootcamp, Automate Cybersecurity Tasks with Python, Developing AI Applications with Python and Flask.
- Other projects: Optical Character Recognition, Automated LinkedIn Job Applier, Classic Pong Game, Classic Snake Game, Turtle Crossing Road Game, Automated Birthday Wisher, Flash Card Game for Learning French.

EDUCATIONAL QUALIFICATIONS

- Pursuing Bachelor's in Computer Science and Engineering from Chandigarh University (Aggregate for 7 Semesters: 7.86 CGPA).
- Completed Class XII From Army Public School, Pune, Maharashtra in Non-Medical with 79.4%.
- Completed Class X from Army Public School, Pune, Maharashtra with 84.5%.

WORK EXPERIENCE

Demandbase

Sep 2024 – Mar 2025

SDE – Cyber security (Intern)

Project: Phishing Simulation

- I created a replica of the frontend of Demandbase's Identity and Access Management (IAM) site. Configured the backend using Python Flask and Zappa to deploy resources (S3 bucket, Lambda function, API Gateway) to AWS and host it in Cloud. API GW endpoint attached to a QR Code, sent to all employees. Approximately 13.4% of all employees scanned the QR code and their data was captured for a knowledge sharing session.

POC: EKS GCP Secure Connectivity

- Established a secure connection between AWS EKS Pod and GCP resources without using Service Account credentials, estimated as 50-80% more secure depending on implementation, due to reduced credential misuse and automatic token expiration.

Module: Slack notifications

- Neglected Resources: Monitored container images/VMs via WIZ, triggering Slack alerts for neglected or outdated resources. Through constant monitoring and compliance our goal was to reduce the count of neglected container images by 10% only but we were able to go well beyond and reduce the count by 30%. For neglected VMs the count was reduced by 8.4%.
- Threat Intel Center: Automated Slack alerts for high profile emerging threats from WIZ's Threat Intel Center, leveraging sources like CISA and CERT-EU, reducing threat response time by 50% and enhancing security awareness for 100% of the team.
- Updates/Blocker: Automated daily standup updates, reducing meetings from 3 to 1 per week, boosting team focus and productivity.

Module: Critical Exposure Issues in Containers

- Developed an automation to identify the container's GitLab repositories, Dockerfiles and pipelines for

compliance, reducing critical exposure issues by 80%.

Module: Secure Terraform modules

- Led shift-left initiative by developing secure Terraform IaC modules for teams, including Lambda-API Gateway, S3, VPC, RDS, EC2, WAF and complex integrations of these modules, following AWS CIS benchmarks. Ensured 100% of resources were securely deployed via Terraform IaC moving forward.

Btech Walleh

Feb 2024 – Apr 2024

SE - Artificial Intelligence Intern

Project: Text Sentiment Analysis ChatBot

- Explored various NLP techniques such as tokenization, word embedding, and built a LSTM neural network model using TensorFlow Keras, Pandas, Numpy, for sentiment classification.
- Trained a model on the imdb_reviews dataset with 50,000+ data points to analyze relationships between text inputs and sentiment labels.
- Conducted extensive testing to ensure the model's performance across a variety of user inputs and sentiments, refining the model iteratively for optimal results, final accuracy came out to be ~75%.

Dabotics India

Feb 2024 – Mar 2024

SDE - Intern

Project: OTP Verification Application

- Crafted a robust OTP verification application using Python, Tkinter, and the SMTPlib module, ensuring ~99% secure user authentication and reducing verification time by 40%.

Project: Conversational ChatBot

- Developed a ChatBot application using Python, enabling interactive conversational experiences through natural language processing and machine learning algorithms, response accuracy was ~70%.

PROJECT EXPERIENCE

Personal Blog Website

Jun 2024 - Jul 2024

- Engineered a user-friendly and fully functional personal blog website where multiple users can post blogs as well as like and comment under others' posts.
- Implemented Bootstrap, HTML, CSS, JavaScript & Jinja templating to construct a beautiful front-end.
- Developed the back-end using SQLite, SQLAlchemy, Python Flask.
- Assembled a Relational DataBase for storing data for users, their posts, and their comments, all 3 tables has relation to each other.
- Administered Hashing and Salting methods to protect user passwords, securing user credentials by ~99%.

RESTful Café API

May 2024 - Jun 2024

- Constructed a RESTful API for a Café database which can be used to get the details of local cafes, post data of a new café into the database, update a café entry in the database as well as delete a café entry from the database given that a secret API-KEY is passed while sending a DELETE request, reducing unauthorized manipulation by ~99%.
- Implemented the REST protocol for API design increasing efficiency by 4-5 times compared to SOAP.
- Assembled an API which can process all 5 requests: GET, POST, PUT/PATCH & DELETE.
- Used Flask for server design along with Flask-SQLAlchemy to fetch data from the SQLite Database, SQLite being 2-3 times faster than MySQL and PostgreSQL for a small scale project.